

Project Proposal

Planning group trips is a common but highly inefficient process, especially among college students and young adults. Despite widespread interest in traveling with friends, many trips never get “out of the group chat” because of logistical challenges. Coordinating schedules, agreeing on budgets, selecting destinations, booking accommodations, and planning activities often require extensive back-and-forth that leads to burnt-out interest. This process is time-consuming, demanding, and falls on one person who becomes responsible for coordinating the entire trip. This problem represents a real-world business opportunity because travel coordination friction directly reduces travel spending and platform engagement.

Roamie is an agentic AI system designed to autonomously coordinate group travel. The system will collect availability, budget constraints, and preferences from participants, identify feasible travel windows, recommend accommodations and activities, facilitate group voting, and complete bookings. By automating the planning workflow, Roamie reduces the burden of coordination and increases the likelihood that trips move from idea to execution.

The primary target users are college students between the ages of 18 and 25, particularly those with social networks across different universities. This demographic frequently attempts to organize trips during school breaks, holidays, and long weekends. Because these students have different academic calendars, financial constraints, and extracurricular commitments, coordinating group travel is especially difficult.

Target users include college students, especially those with friends across different universities. It is a common occurrence for the average student to have friends across different colleges, as their high school friends may not have gone to the same college as them. Maintaining those friendships takes much effort, and one form of this is planning a trip. Whether it be between old friends or new, coordinating between a group of friends is always difficult. More of the target audience includes: young professionals planning trips with friends in different cities, friend groups planning birthday celebrations or reunions, student organizations planning group outings, and small social groups planning local hangouts such as dinners or day trips. These users share a common need: reducing the coordination burden in group events.

An agentic AI solution is particularly well-suited for this problem because trip planning is a multi-step, goal-oriented workflow that requires autonomous decision-making, tool usage, and iterative coordination. Students are busy with college and other extracurricular endeavors. An agentic AI solution is appropriate, as it wouldn't require human intervention. The student can just input the availability of the people in the group, and the AI will take it from there.

Scenario #1: Spring Break Trip Planning

A user wants to do a group trip for Spring Break.

The AI will:

- Take the group's availability and the length of the trip
- Identify overlapping availability
- Collect individual budget ranges
- Recommend destinations within budget and travel distance constraints
- Search for accommodations
- Present options and facilitate group voting
- Book the selected accommodation (the one with the majority of votes)
- Recommend activities such as restaurants and attractions in the vicinity of the accommodation
- Create and distribute a shared itinerary

Scenario #2: Birthday Dinner Planning

Suppose a user just wants to plan a quick hangout, like a lunch or dinner, for someone's birthday.

The AI will

- Take the group's availability in dates and times
- Identify a suitable date and time
- Search for restaurants matching cuisine and budget preferences
- Make reservations
- Send confirmations and reminders to attendees

Scenario #3: Weekend Reunion Trip

A group of friends living in different cities wants to plan a weekend reunion.

The AI will:

- Analyze travel distances to suggest a central meeting location
- Recommend accommodations
- Suggest activities based on shared interests
- Handle booking and itinerary creation

Initial system design

The system will have a Planner Agent, Accommodation Agent, Reservation Agent, Voting Agent, Activity Agent, and Communicator Agent.

Planner Agent:

- takes the available dates of each attendee
- selects a suitable time frame
- also takes in the length of the event (i.e., 2 days, overnight trip; 2-3 hour dinner)

Accommodation Agent:

- inputs the location/city the group wants to visit
- Inputs the budget
- sends a list of recommendations to each attendee's email
- Each user needs to vote and pick the accommodation they like the most. The place with the highest votes will get booked.

Reservation Agent:

- Gets the time for the reservation, the cuisine, and the general location from the planner
- Make restaurant reservations
- Send confirmations

Activity Agent:

- Search for local attractions and activities
- Recommend options based on user interests (if provided)
- Send to the Communication Agent to build the itinerary

Communication Agent:

- Send updates via email or messaging platforms
- Notify users of decisions
- Send reminders
- Build itinerary

Voting Agent:

- Collect user votes via email
- Identify winning options - mostly useful for the Accommodation Agent
- Resolve conflicts

Expected tools/APIs or data sources

Accommodation:

- Expedia API [to look for accommodations]
- Gmail API [to send the possible accommodations to everyone]

Activity:

- Google Maps API [to look for attractions]
- Yelp API [to look for attractions]
- Summarize tool to send items to the Communication Agent

Communication:

- Gmail API [to send the email updates]
- Google Calendar API [to help the group add the dates to their calendar]
- LLM tool to build an itinerary

Reservation:

- Google Maps API
- OpenTable API

Voting:

- Google Forms [to take in the vote]
- Google Sheets [where the LLM will find the stored votes]
- LLM [to read the votes and select the majority option]

Feasibility & risks

Potential challenges:

- Hallucination [the Agents may select an unavailable or incorrect accommodation]
- Complexity due to multiple agents
- Tool Failures [external APIs may fail]

Any constraints:

- OpenTable API is not available to the public, so the Reservation Agent may just become a Recommendation Agent
- The AI will need to use sensitive information like availability schedules and email addresses, so the stored data needs to be secured
- Need to use efficient queries to minimize usage costs and stay within the rate limits

Time limit for project execution:

- Week 1: Create a basic layout of how the agents will interact with each other, and decide on a parallel, loop, or sequential path(s).
- Week 2: Build the user-interface
- Week 3: Create the voter system, activity and accommodations systems
- Week 4: Integrate the external APIs
- Week 5: Test and Debug

At the moment, the timeline listed is not accurate and will need to be discussed with the professor to match reality and gain a better understanding of what order this needs to be accomplished in.

Statement of AI Use:

ChatGPT was used in this assignment to discuss the feasibility of this AI in an Agentic implementation, discuss possible APIs to use, and to better summarize and convey the idea.